SSC CGL 2024 Tier-I question paper with answers

24 September 2024 · Shift 1

81 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: +2 for a right answer, –0.50 for a wrong one

The other 19 of the paper's 100 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/Writeup_Final_Anwerkey_CGLE_2024_T1_191224.pdf

Compare your score: the official 2024 cut-off (Tier-I, all posts except JSO and SI Grade-II), out of 200

UR 153.19 **OBC** 146.26 **EWS** 142.02 **SC** 126.46 **ST** 111.89 **ESM** 69.93 **OH** 113.10 **HH** 64.79 **VH** 102.97

Other PwD 45.74

Cut-offs apply to marks normalised across all the exam's shifts, so a raw score on one shift is a rough guide. Source: the official result write-up, https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Result%20Write-up%20CGLE%202024%20Tier%20I_51224.pdf

General Intelligence and Reasoning 18 questions

Q.1

'RX-21' is related to 'UU-11' in a certain way. In the same way, 'PM-39' is related to 'SJ-29'. To which of the following is DP-17 related, following the same logic?

(A) GM-10
(B) HN-7
(C) GM-7
(D) HO-10

Q.2

Select the triad in which the numbers are related in the same way as are the numbers of the given triads. (8, 9, 7), (24, 13, 35) (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)

(A) (21, 12, 32)
(B) (11, 15, 19)
(C) (28, 19, 56)
(D) (17, 11, 23)

Q.3

Select the number from the given options to complete the series. 32, 36, 44, 56, 72, ___

(A) 98
(B) 92
(C) 80
(D) 90

Q.4

Three of the following four options are alike in a certain way and thus form a group. Which is the one that does NOT belong to that group? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 58 – 139 – 23
(B) 23 – 84 – 38
(C) 89 – 212 – 34
(D) 12 – 122 – 99

Q.5

61 is related to 18 following a certain logic. Following the same logic, 213 is related to 38. To which of the following is 509 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 78
(B) 64
(C) 72
(D) 66

Q.6

In a certain code language, 'LIGHT' is coded as '25709' and 'SIGHT' is coded as '97065'. What is the code for 'S' in that language?

(A) 2
(B) 7
(C) 6
(D) 0

Q.7

Select the option that is related to the fifth letter-cluster in the same way as the second letter-cluster is related to the first letter-cluster and the fourth letter-cluster is related to the third letter-cluster. LOI: QTN :: VES:AJX :: AND: ?

(A) FSJ
(B) FSI
(C) FTJ
(D) FTI

Q.10

The position of how many letters will remain unchanged if each of the letters in the word 'DELIGHT' is arranged in the English alphabetical order?

(A) One
(B) None
(C) Three
(D) Two

Q.11

If A is the sister of B, C is the daughter of A, D is the sister of A and E is the brother of C. How is D related to E?

(A) Mother's mother
(B) Mother's father
(C) Mother's brother
(D) Mother's sister

Q.12

In a certain code language, 'PUPIL' is written as '162116912' and 'PURGE' is written as '16211875'. How will 'PURSE' be written in that language?

(A) 162118159
(B) 126118195
(C) 162111895
(D) 162118195

Q.14

What will come in place of the question mark (?) in the following equation, if '+' and '÷' are interchanged and '×' and '–' are interchanged? 20 + 4 ÷ 2 × 3 – 6 = ?

(A) –11
(B) –7
(C) 2
(D) –9

Q.15

Select the combination of letters that when sequentially placed in the blanks of the given series will complete the series. _TUVTR_WU_UXVN_Y

(A) SUUP
(B) SUPU
(C) PSVU
(D) SVUP

Q.16

Select the option in which the numbers share the same relationship as that shared by the given pair of numbers. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/deleting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 33 : 340 48 : 490

(A) 27 : 270
(B) 31 : 279
(C) 44 : 460
(D) 63 : 640

Q.17

Three statements are given, followed by three conclusions numbered I, II and III. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusions logically follow(s) from the statements. Statements: All spoons are forks. All forks are knives. No knife is a spatula. Conclusions: I. Some knives are spoons. II. No fork is a spatula. III. Some spoons are spatulas.

(A) Neither conclusion follows
(B) Only conclusion III follows
(C) Only conclusions I and II follow
(D) Only conclusion I follows

Q.22

Which of the following letter-clusters will replace the question mark (?) in the given series to make it logically complete? TGO, VIN, XKM, ZML, ?

(A) BOM
(B) BNN
(C) XKO
(D) BOK

Q.23

What will come in place of the question mark (?) in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 125 ÷ 5 + 1809 × 9 – 14 = ?

(A) 348
(B) 438
(C) 483
(D) 384

Q.24

Four letter-clusters have been given, out of which three are alike in some manner and one is different. Select the one that is different. The odd one out is not based on the number of consonants/vowels or their position in the letter cluster

(A) UTW
(B) LKM
(C) NMP
(D) CBE

Q.25

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some Mirrors are Decoratives. No Decorative is a Plant. All Plants are Trees. Conclusions: (I) Some Mirrors are Plants. (II) Some Decoratives are Trees.

(A) Neither conclusion (I) nor (II) follows.
(B) Only conclusion (I) follows.
(C) Both conclusions (I) and (II) follow.
(D) Only conclusion (II) follows.

General Awareness 25 questions

Q.1

Who founded the Arya Samaj in Mumbai in 1875?

(A) Swami Vivekanand
(B) Swami Dayanand Saraswati
(C) Raja Ram Mohan Roy
(D) Debendranath Tagore

Q.2
Who is considered the highest law officer of the state in India?

(A) Auditor general
(B) Advocate general
(C) Chief justice of high court
(D) Attorney general

Q.3
In which state of India is the Tadoba National Park located?

(A) Madhya Pradesh
(B) Kerala
(C) Gujarat
(D) Maharashtra

Q.4
What is the primary benefit offered to businesses in Special Economic Zones (SEZs)?

(A) Tax and duty concessions
(B) Mandatory government contracts
(C) Guaranteed market share
(D) Unlimited foreign investment

Q.5
Which of the following statements is correct?

(A) Rivers and sea water are considered as fresh water.
(B) Pond water and sea water are considered as fresh water.
(C) Rivers, creeks, lakes, ponds, and streams are all freshwater habitats.
(D) Sea water is freshwater habitat.

Q.6
Which of the following countries will host the ICC men's Cricket World Cup in the year 2031?

(A) Sri Lanka and Pakistan
(B) India, Bangladesh
(C) Sri Lanka and Bangladesh
(D) India, Sri Lanka

Q.7
Which of the following terms is related to football?

(A) Bicycle Kick
(B) LBW
(C) Overs
(D) Third man

Q.8
AR Rehman, Gulzar and _____ won the 52nd annual Grammy Award for the best motion picture song 'Jai ho'.

(A) Danny Boyle
(B) Dev Patel
(C) Tanvi Shah
(D) Freida Pinto

Q.9
The Working Group under the chairmanship of _____ in the year_____ proposed a new intermediate monetary aggregate to be referred to as NM2.

(A) Dr. C Rangarajan; 1996
(B) Dr. KV Kamath; 1995
(C) Dr. PK Mohanty; 1998
(D) Dr. YV Reddy; 1998

Q.10
Classical Sarod player, Amjad Ali Khan was awarded _____ in 2001 by the Government of India.

(A) Padma Vibhushan
(B) Padma Shri
(C) Bharat Ratna
(D) Padma Bhushan

Q.11
By which National Waterways (NW) of India, are the delta channels of Mahanadi and Brahmani rivers and East Coast Canal connected?

(A) NW 1
(B) NW 2
(C) NW 5
(D) NW 3

Q.12
According to the Census of India 2011, which group of Union Territories has the highest density of population?

(A) Delhi, Chandigarh and Daman and Diu
(B) Lakshadweep, Daman and Diu and Puducherry
(C) Delhi, Chandigarh and Puducherry
(D) Lakshadweep, Daman and Diu and Chandigarh

Q.13
Identify an example of plasma as a state of matter.

(A) Neon sign bulbs
(B) Freon
(C) Blood
(D) Dry ice

Q.14
Which fundamental duty applies to you when you see your college friends start fighting with the canteen person when he passed a few anti-religious statements?

(A) To value and preserve the rich heritage
(B) To promote harmony and spirit of brotherhood
(C) To protect and improve our natural environment
(D) To develop the scientific temper, humanism, and the spirit of inquiry

Q.15
Which Article deals with the election of the President?

(A) Article 66
(B) Article 54
(C) Article 74
(D) Article 72

Q.16
Who among the following Indian cricketers were named the Wisden's cricketer of the year in the 2022 edition of the Almanack?

(A) Rohit Sharma and Virat Kohli
(B) Surya Kumar Yadav and Axar Patel
(C) Virat Kohli and Bhuvneshwar Kumar
(D) Rohit Sharma and Jasprit Bumrah

Q.17
Which of the following is NOT a physical change?

(A) Curdling of milk
(B) Heating of iron rod to red hot
(C) Evaporation of diesel
(D) Sublimation of NH_4Cl

Q.18
Who is called the father of Local Self-Government in India?

(A) Lord Cornwallis
(B) Lord Wellesley
(C) Lord Ripon
(D) Lord William Bentinck

Q.19
By which of the following methods do red algae reproduce?

(A) Grafting
(B) Cutting
(C) Fragmentation
(D) Micropropagation

Q.20

By what name is the festival of Makar Sankranti celebrated in Assam?

(A) Pongal
(B) Thai Pongal
(C) Magh Bihu
(D) Pedda Padunga

Q.21

Which of the following Constitution Amendment Bills is related to the provision of modifying the list of Scheduled Caste in Tamil Nadu?

(A) The Constitution (Scheduled Tribes) Order Amendment Act, 2013
(B) The Constitution (Scheduled Castes) Order (Amendment) Bill, 2021
(C) The Constitution (Scheduled Tribes) Order (Amendment) Bill, 2019
(D) The Constitution (Scheduled Caste) Order (Amendment) Bill, 2015

Q.22

Mahendravarman I who wrote Mattavilasa-prahasana is related to which of the following ruling dynasties?

(A) Pushyabhuti
(B) Chalukya
(C) Maitraka
(D) Pallava

Q.23

When was the All India Trade Union Congress (AITUC) established?

(A) 1920
(B) 1926
(C) 1924
(D) 1928

Q.24

Which of the following is responsible for the red colour of beetroot?

(A) Lycopene
(B) Betanin
(C) Curcumin
(D) Beta carotene

Q.25

Raziya Sultan was the first and only female ruler of the Sultanate, who ascended the throne in:

(A) 1238 AD
(B) 1242 AD
(C) 1241 AD
(D) 1236 AD

Quantitative Aptitude 13 questions

Q.1

Find the greatest value of ‘k’ in a 6-digit number 6745k2 such that the number is divisible by 3.

(A) 7
(B) 8
(C) 6
(D) 9

Q.3

One pipe can fill a tank 6 times faster than another pipe. If both the pipes together can fill the tank in 40 minutes, then in how many minutes will the slower pipe alone be able to fill the tank?

(A) 290
(B) 275
(C) 285
(D) 280

Q.5

What conclusion can be drawn about the solution of the following system of linear equations in two variables: $3x + 2y = 7$ $2x + 3y = 7$

(A) More than two solution
(B) Infinite solutions
(C) Unique solution
(D) No solution

Q.8

The marked price of a revolving chair is ₹15,000. It is available on two successive discounts of 20% and 12%. What is its selling price (in ₹)?

(A) 9,560
(B) 10,230
(C) 9,230
(D) 10,560

Q.10

The radius of a circle is 12.5 cm and the length of one its chords is 11 cm. What is the distance of the chord from the centre? (correct to one decimal place)

(A) 10 cm
(B) 12 cm
(C) 11.2 cm
(D) 13.2 cm

Q.12

In an election between two candidates, 80% of the voters cast their votes, out of which 4% votes were declared invalid. A candidate got 15,360 votes which were 80% of the valid votes. Find the total number of votes.

(A) 18,000
(B) 23,000
(C) 25,000
(D) 20,000

Q.13

PQ is the diameter of a circle with centre A. M is a point on the circumference of the circle. If $m\angle MAP = 110°$, then find the value of $m\angle MQA$.

(A) 70°
(B) 55°
(C) 35°
(D) 65°

Q.14

The volume of a right circular cylindrical box of radius 30cm is 600cm^3. Find the height of the box.

(A) 0.21cm
(B) 0.61cm
(C) 0.51cm
(D) 0.31cm

Q.15

Triangle XYZ is an isosceles triangle with sides XY = XZ. If the measure of angle Y is 80°, what is the measure of angle X?

(A) 30°
(B) 40°
(C) 50°
(D) 20°

Q.17

In how much time (in years, rounded off to 2 decimal places) will ₹10,800 amount to ₹24,800 if invested at simple interest at the rate of 15.5% per annum?

(A) 7.72
(B) 9.54
(C) 6.89
(D) 8.36

Q.18

In what ratio, sugar costing ₹78 per kg be mixed with sugar costing ₹46 per kg such that by selling the mixture at ₹75 per kg there is a gain of 20%?

(A) 37 : 31
(B) 33 : 31
(C) 31 : 29
(D) 33 : 37

Q.23

A thief is spotted by a policeman from a distance of 100 m. The thief starts running and the policeman chases him. If the speed of thief and policeman are 21 km/h and 23 km/h, respectively, then how far will the thief have to run before he is over taken?

(A) 1090 m
(B) 1050 m
(C) 1080 m
(D) 1020 m

Q.24

A, B and C invested money in a business such that 2 times A's share, 4 times B's share and 6 times C's share are all equal. If they earned ₹44,000, find the share of A.

(A) ₹12,000
(B) ₹22,000
(C) ₹24,000
(D) ₹20,000

English Comprehension 25 questions

Q.1

Select the most appropriate phrasal verb to fill in the blank. We need to _____ a solution to this problem before it gets worse.

(A) cut down on
(B) break down
(C) come up with
(D) break up

Q.2

Select the most appropriate idiom for the given group of words. A time when money might unexpectedly be needed

(A) A rash of something
(B) A raw deal
(C) A rap over the knuckles
(D) A rainy day

Q.3

Select the most appropriate option that can substitute the underlined segment in the given sentence. Mr. Adhil “called back but he wasn’t able to” find the scooter part he needed.

(A) called down but he wasn’t able to
(B) called up to but he wasn’t able to
(C) called off but he wasn’t able to
(D) called around but he wasn’t able to

Q.4

Select the most appropriate ANTONYM of the given word from the following sentence. Barren "Can you believe how much traffic there is today?", grumbled Tom, glancing at the congested road ahead.

(A) Grumbled
(B) Glancing
(C) Believe
(D) Congested

Q.5

Select the most appropriate ANTONYM of the underlined word in the following sentence. It was difficult to accommodate the “avariciousness” of the people.

(A) Criticism
(B) Envy
(C) Hatred
(D) Generosity

Q.6

Select the INCORRECTLY spelt word..

(A) Confidential
(B) Measerement
(C) Region
(D) Strengths

Q.7

Select the most appropriate option to substitute the underlined word segment in the given sentence. A “herd of cattle” consists of related females, cubs, and a small number of adult males.

(A) pride of lions
(B) school of fish
(C) flock of birds
(D) pack of wolves

Q.8

Select the most appropriate synonym of the given word. Maverick

(A) Conformist
(B) Bohemian
(C) Orthodox
(D) Follower

Q.9

Identify the error in the given sentence. Camels tend to move slowly than other domestic animals in the plains.

(A) domestic animals in the plains.
(B) move slowly
(C) Camels tend to
(D) than other

Q.10

Select the option that can be used as a one-word substitute for the given group of words. A person who does not believe in God

(A) Atheist
(B) Amateur
(C) Anarchy
(D) Autocrat

Q.11

Select the most appropriate option that can substitute the underlined word in the given sentence. I saw a shabby looking man living in the “cavity” of a mountain.

(A) height
(B) elevation
(C) mound
(D) hole

Q.12

Select the most appropriate synonym of the underlined word in the given sentence. This is a clear example of “conflict” of egos.

(A) mingling
(B) infliction
(C) clash
(D) crash

Q.13

Parts of a sentence are given below in jumbled order. Arrange the parts in the correct order to form a meaningful sentence. P) her face Q) I have R) for a year S) not seen

(A) SPQR
(B) PQRS
(C) QSPR
(D) PSQR

Q.14

Select the most appropriate meaning of the underlined idiom in the following sentence. Instead of confusing others, let us “call a spade a spade” in front of the audience.

(A) Be at strife
(B) Remain silent
(C) Pretend superiority
(D) Speak truthfully

Q.15

Select the INCORRECTLY spelt word.

(A) Connoisseur
(B) Privilege
(C) Neccessary
(D) Dilemma

Q.16

Select the most appropriate option that can substitute the underlined segment in the given sentence. It is nothing “else your pride which makes” you say such a thing.

(A) else but your pride which make
(B) else your pride that makes
(C) else but your pride which makes
(D) else your pride which make

Q.17

Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph. A.The door of the flat was ajar. B.The taxi stopped. C.Mr. Satterthwaite flung himself out and raced up the stone stairs to the second floor like a young athlete. D.He pushed it open, and the great voice welcomed him.

(A) ABCD
(B) BCAD
(C) DCBA
(D) BACD

Q.18

Select the option that expresses the given sentence in passive voice. He has written a best-selling novel.

(A) A best-selling novel has been written by him.
(B) A best-selling novel was written by him.
(C) A best-selling novel was being written by him
(D) A best-selling novel would have been written by him.

Q.19

Select the option that expresses the given sentence in passive voice. He wrote an inspiring speech for the event.

(A) An inspiring speech was written by him for the event.
(B) An inspiring speech was being written by him for the event.
(C) An inspiring speech has been written by him for the event.
(D) An inspiring speech is written by him for the event.

Q.20

Select the most appropriate meaning of the underlined word in the given sentence. This building is “colossal” and it can hold a village’s population in it.

(A) very beautiful
(B) very huge
(C) puny
(D) temporary

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The mathematician Federico Ardila-Mantilla grew up in Colombia, an (1) _____ student but gifted in math. He was failing most of his classes at his high school in Bogotá when someone (2) _____ him to apply to MIT. He had not heard of the school. To his surprise, he got in, and he went on scholarship. Mathematically, he did well. One of his professors, an acid-tongued theoretician known to (3) _____ his audience to a herd of cows, routinely tucked 'open' math problems into homework assignments, without telling the students. These had never been solved by anyone. Ardila solved one. He went on to receive his bachelor's and Ph.D. in math from MIT. But his academic experience was also one of isolation. Part of it had to do with his own (4) _____. Part of it was cultural. As a Latino, he was very much in the minority in the department, and he did not feel comfortable in American mathematical spaces. No one had tried to explicitly (5) _____ him, yet he felt alone. In math, collaborating with others opens up new kinds of learning and thinking. But in his nine years at MIT, Ardila worked with others only twice.

Select the most appropriate option to fill in blank no. 1

(A) indifferent
(B) apprehensive
(C) apologetic
(D) excellent

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The mathematician Federico Ardila-Mantilla grew up in Colombia, an (1) _____ student but gifted in math. He was failing most of his classes at his high school in Bogotá when someone (2) _____ him to apply to MIT. He had not heard of the school. To his surprise, he got in, and he went on scholarship. Mathematically, he did well. One of his professors, an acid-tongued theoretician known to (3) _____ his audience to a herd of cows, routinely tucked 'open' math problems into homework assignments, without telling the students. These had never been solved by anyone. Ardila solved one. He went on to receive his bachelor's and Ph.D. in math from MIT. But his academic experience was also one of isolation. Part of it had to do with his own (4) _____. Part of it was cultural. As a Latino, he was very much in the minority in the department, and he did not feel comfortable in American mathematical spaces. No one had tried to explicitly (5) _____ him, yet he felt alone. In math, collaborating with others opens up new kinds of learning and thinking. But in his nine years at MIT, Ardila worked with others only twice.

Select the most appropriate option to fill in blank no. 2

(A) dissuaded
(B) suggested
(C) dispirited
(D) removed

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The mathematician Federico Ardila-Mantilla grew up in Colombia, an (1) _____ student but gifted in math. He was failing most of his classes at his high school in Bogotá when someone (2) _____ him to apply to MIT. He had not heard of the school. To his surprise, he got in, and he went on scholarship. Mathematically, he did well. One of his professors, an acid-tongued theoretician known to (3) _____ his audience to a herd of cows, routinely tucked 'open' math problems into homework assignments, without telling the students. These had never been solved by anyone. Ardila solved one. He went on to receive his bachelor's and Ph.D. in math from MIT. But his academic experience was also one of isolation. Part of it had to do with his own (4) _____. Part of it was cultural. As a Latino, he was very much in the minority in the department, and he did not feel comfortable in American mathematical spaces. No one had tried to explicitly (5) _____ him, yet he felt alone. In math, collaborating with others opens up new kinds of learning and thinking. But in his nine years at MIT, Ardila worked with others only twice.

Select the most appropriate option to fill in blank no. 3

(A) compare
(B) differentiate
(C) strangulate
(D) malign

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The mathematician Federico Ardila-Mantilla grew up in Colombia, an (1) _____ student but gifted in math. He was failing most of his classes at his high school in Bogotá when someone (2) _____ him to apply to MIT. He had not heard of the school. To his surprise, he got in, and he went on scholarship. Mathematically, he did well. One of his professors, an acid-tongued theoretician known to (3) _____ his audience to a herd of cows, routinely tucked 'open' math problems into homework assignments, without telling the students. These had never been solved by anyone. Ardila solved one. He went on to receive his bachelor's and Ph.D. in math from MIT. But his academic experience was also one of isolation. Part of it had to do with his own (4) _____. Part of it was cultural. As a Latino, he was very much in the minority in the department, and he did not feel comfortable in American mathematical spaces. No one had tried to explicitly (5) _____ him, yet he felt alone. In math, collaborating with others opens up new kinds of learning and thinking. But in his nine years at MIT, Ardila worked with others only twice.

Select the most appropriate option to fill in blank no. 4.

(A) quantification
(B) introversion
(C) interrogation
(D) molestation

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. The mathematician Federico Ardila-Mantilla grew up in Colombia, an (1) _____ student but gifted in math. He was failing most of his classes at his high school in Bogotá when someone (2) _____ him to apply to MIT. He had not heard of the school. To his surprise, he got in, and he went on scholarship. Mathematically, he did well. One of his professors, an acid-tongued theoretician known to (3) _____ his audience to a herd of cows, routinely tucked 'open' math problems into homework assignments, without telling the students. These had never been solved by anyone. Ardila solved one. He went on to receive his bachelor's and Ph.D. in math from MIT. But his academic experience was also one of isolation. Part of it had to do with his own (4) _____. Part of it was cultural. As a Latino, he was very much in the minority in the department, and he did not feel comfortable in American mathematical spaces. No one had tried to explicitly (5) _____ him, yet he felt alone. In math, collaborating with others opens up new kinds of learning and thinking. But in his nine years at MIT, Ardila worked with others only twice.

Select the most appropriate option to fill in blank no. 5.

(A) fabricate
(B) exclude
(C) incorporate
(D) stratify

Answer key

General Intelligence and Reasoning

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(D)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(B)**	Q.10 **(C)**	Q.11 **(D)**	Q.12 **(D)**	Q.14 **(A)**	Q.15 **(B)**
Q.16 **(D)**	Q.17 **(C)**	Q.22 **(D)**	Q.23 **(B)**	Q.24 **(B)**	Q.25 **(A)**

General Awareness

Q.1 **(B)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(B)**
Q.7 **(A)**	Q.8 **(C)**	Q.9 **(D)**	Q.10 **(A)**	Q.11 **(C)**	Q.12 **(C)**
Q.13 **(A)**	Q.14 **(B)**	Q.15 **(B)**	Q.16 **(D)**	Q.17 **(A)**	Q.18 **(C)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(B)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(D)**					

Quantitative Aptitude

Q.1 **(D)**	Q.3 **(D)**	Q.5 **(C)**	Q.8 **(D)**	Q.10 **(C)**	Q.12 **(C)**
Q.13 **(B)**	Q.14 **(A)**	Q.15 **(D)**	Q.17 **(D)**	Q.18 **(B)**	Q.23 **(B)**
Q.24 **(C)**					

English Comprehension

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(D)**	Q.5 **(D)**	Q.6 **(B)**
Q.7 **(A)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(A)**	Q.11 **(D)**	Q.12 **(C)**
Q.13 **(C)**	Q.14 **(D)**	Q.15 **(C)**	Q.16 **(C)**	Q.17 **(B)**	Q.18 **(A)**
Q.19 **(A)**	Q.20 **(B)**	Q.21 **(A)**	Q.22 **(B)**	Q.23 **(A)**	Q.24 **(B)**
Q.25 **(B)**					

Explanations

General Intelligence and Reasoning

Q.1 · Answer (C) GM-7

The first letter moves 3 forward, the second 3 back and the number falls by 10: RX-21 → UU-11 and PM-39 → SJ-29. So DP-17 → GM-7.

Q.2 · Answer (D) (17, 11, 23)

In each triad, twice the first number minus the second gives the third: 2 × 8 – 9 = 7 and 2 × 24 – 13 = 35. Likewise 2 × 17 – 11 = 23.

Q.3 · Answer (B) 92

The differences are 4, 8, 12, 16, rising by 4 each time, so the next is 20: 72 + 20 = 92.

Q.4 · Answer (D) 12 – 122 – 99

In the others, twice the first number plus the third gives the middle one: 2 × 58 + 23 = 139, 2 × 23 + 38 = 84 and 2 × 89 + 34 = 212. But 2 × 12 + 99 = 123, not 122.

Q.5 · Answer (D) 66

Each first number is a cube minus 3, and the second is the square of the same number plus 2: 61 = 4^3 – 3 gives 4^2 + 2 = 18, and 213 = 6^3 – 3 gives 6^2 + 2 = 38. 509 = 8^3 – 3, so 8^2 + 2 = 66.

Q.6 · Answer (C) 6

I, G, H and T are in both words, as are 0, 5, 7 and 9 in both codes. The letter left in SIGHT is S and the digit left in its code is 6, so S = 6.

Q.7 · Answer (B) FSI

Each letter moves 5 places forward, wrapping round after Z: LOI → QTN and VES → AJX. So AND → FSI.

Q.10 · Answer (C) Three

DELIGHT in alphabetical order is D E G H I L T. D, E and T stay in the same places: three letters.

Q.11 · Answer (D) Mother's sister

E is C's brother, so C's mother A is E's mother too. D is A's sister, so D is E's mother's sister.

Q.12 · Answer (D) 162118195

Each letter is written as its place in the alphabet: P = 16, U = 21, R = 18, S = 19, E = 5. So PURSE is 162118195.

Q.14 · Answer (A) –11

After interchanging the signs: 20 ÷ 4 + 2 – 3 × 6 = 5 + 2 – 18 = –11.

Q.15 · Answer (B) SUPU

With S, U, P, U in the blanks the series reads STUV TRUW UPUX VNUY: the first letters run S, T, U, V, the second T, R, P, N (back 2), the third is always U and the fourth runs V, W, X, Y.

Q.16 · Answer (D) 63 : 640

The second number is ten times the first plus 10: 33 × 10 + 10 = 340 and 48 × 10 + 10 = 490. Likewise 63 × 10 + 10 = 640.

Q.17 · Answer (C) Only conclusions I and II follow

All spoons are forks and all forks are knives, so all spoons are knives and some knives are spoons (I). No knife is a spatula, so no fork is a spatula (II), and no spoon is one either, so III does not follow.

Q.22 · Answer (D) BOK

First letters move 2 forward (T, V, X, Z, B), second letters 2 forward (G, I, K, M, O) and third letters 1 back (O, N, M, L, K): BOK.

Q.23 · Answer (B) 438

After interchanging the signs: 125 × 5 – 1809 ÷ 9 + 14 = 625 – 201 + 14 = 438.

Q.24 · Answer (B) LKM
In UTW, NMP and CBE the letters move 1 back, then 3 forward. LKM moves 1 back, then 2 forward, so it is the odd one.

Q.25 · Answer (A) Neither conclusion (I) nor (II) follows.
The mirrors are linked to plants only through decoratives, and no decorative is a plant, so I need not follow. Nothing says whether any decorative is a tree, so II need not follow either.

General Awareness

Q.1 · Answer (B) Swami Dayanand Saraswati
Swami Dayanand Saraswati founded the Arya Samaj in Bombay in 1875, calling for a return to the Vedas.

Q.2 · Answer (B) Advocate general
Article 165 provides for an Advocate General in each state, appointed by the Governor, as the state's highest law officer. The Attorney General (Article 76) is the highest law officer of the Union.

Q.3 · Answer (D) Maharashtra
Tadoba National Park, part of the Tadoba-Andhari Tiger Reserve, is in Chandrapur district, Maharashtra.

Q.4 · Answer (A) Tax and duty concessions
SEZs offer their units tax and duty concessions, such as duty-free imports and income-tax breaks, to promote exports.

Q.5 · Answer (C) Rivers, creeks, lakes, ponds, and streams are all freshwater habitats.
Rivers, creeks, lakes, ponds and streams hold fresh water; sea water is salt water, a marine habitat.

Q.6 · Answer (B) India, Bangladesh
In November 2021 the ICC named India and Bangladesh joint hosts of the 2031 Men's Cricket World Cup.

Q.7 · Answer (A) Bicycle Kick
A bicycle (overhead) kick is a football move; LBW, overs and third man belong to cricket.

Q.8 · Answer (C) Tanvi Shah
Tanvi Shah, who wrote and sang the Spanish lines of 'Jai Ho', shared the Grammy with A R Rahman and Gulzar.

Q.9 · Answer (D) Dr. YV Reddy; 1998
The RBI's Working Group on Money Supply chaired by Dr Y V Reddy reported in 1998; it proposed new monetary aggregates, with NM2 as the intermediate one.

Q.10 · Answer (A) Padma Vibhushan
Sarod maestro Amjad Ali Khan received the Padma Vibhushan, India's second-highest civilian award, in 2001.

Q.11 · Answer (C) NW 5
National Waterway 5 joins the East Coast Canal with the Brahmani-Kharsua river system and the delta channels of the Mahanadi, in Odisha and West Bengal.

Q.12 · Answer (C) Delhi, Chandigarh and Puducherry
In the 2011 census Delhi, Chandigarh and Puducherry were the three most densely populated Union Territories.

Q.13 · Answer (A) Neon sign bulbs
In a neon sign an electric current ionises the gas into a glowing plasma. Freon is a gas, blood a liquid and dry ice a solid.

Q.14 · Answer (B) To promote harmony and spirit of brotherhood
Article 51A(e) asks every citizen to promote harmony and the spirit of common brotherhood, transcending religious, linguistic and regional diversities.

Q.15 · Answer (B) Article 54
Article 54 provides for the election of the President by an electoral college; Article 66 covers the Vice-President's election.

Q.16 · Answer (D) Rohit Sharma and Jasprit Bumrah
Rohit Sharma and Jasprit Bumrah were among Wisden's five Cricketers of the Year in the 2022 Almanack.

Q.17 · Answer (A) Curdling of milk
Curdling of milk is a chemical change: acid coagulates the milk's proteins, and the change cannot be reversed. The other three change only the temperature or state of the substance.

Q.18 · Answer (C) Lord Ripon
Lord Ripon's resolution of 1882 laid the foundation of local self-government in India.

Q.19 · Answer (C) Fragmentation
Red algae usually reproduce vegetatively by fragmentation; they also reproduce by spores and sexually.

Q.20 · Answer (C) Magh Bihu
In Assam Makar Sankranti is celebrated as Magh (Bhogali) Bihu, the harvest festival; Pongal is Tamil Nadu's name for it.

Q.21 · Answer (B) The Constitution (Scheduled Castes) Order (Amendment) Bill, 2021
The Bill modified Tamil Nadu's list of Scheduled Castes, grouping seven castes under the common name Devendrakula Velalar.

Q.22 · Answer (D) Pallava
Mahendravarman I, the Pallava king of Kanchi, wrote the Sanskrit farce 'Mattavilasa Prahasana'.

Q.23 · Answer (A) 1920
The AITUC was founded in Bombay on 31 October 1920, with Lala Lajpat Rai as its first president.

Q.24 · Answer (B) Betanin
Betanin, a betalain pigment, gives beetroot its red colour. Lycopene colours tomatoes, curcumin turmeric and beta-carotene carrots.

Q.25 · Answer (D) 1236 AD
Iltutmish's daughter Raziya ascended the throne of Delhi in 1236 and ruled until 1240.

Quantitative Aptitude

Q.1 · Answer (D) 9
The digit sum 6 + 7 + 4 + 5 + k + 2 = 24 + k must be a multiple of 3, so k is 0, 3, 6 or 9. The greatest is 9.

Q.3 · Answer (D) 280
If the slower pipe fills 1 part a minute, the faster fills 6, together 7. The tank holds 7 × 40 = 280 parts, so the slower pipe alone takes 280 minutes.

Q.5 · Answer (C) Unique solution
3/2 ≠ 2/3, so the two lines cross at exactly one point: the unique solution x = 1, y = 1.

Q.8 · Answer (D) 10,560
₹15,000 × 0.8 × 0.88 = ₹10,560.

Q.10 · Answer (C) 11.2 cm
Distance = $\sqrt{(r^2 - (\text{half the chord})^2)} = \sqrt{(12.5^2 - 5.5^2)} = \sqrt{(156.25 - 30.25)} = \sqrt{126} \approx 11.2$ cm.

Q.12 · Answer (C) 25,000
Valid votes = 15,360/0.8 = 19,200, which is 96% of the votes cast: 20,000. These are 80% of the voters, so there were 25,000 votes in all.

Q.13 · Answer (B) 55°
$\angle MAQ = 180° - 110° = 70°$. AM = AQ (radii), so $\angle MQA = (180° - 70°)/2 = 55°$.

Q.14 · Answer (A) 0.21cm
$\pi r^2 h = 600$, so $h = 600/(\pi \times 900) = 2/(3\pi) \approx 0.21$ cm.

Q.15 · Answer (D) 20°
As XY = XZ, $\angle Z = \angle Y = 80°$, so $\angle X = 180° - 160° = 20°$.

Q.17 · Answer (D) 8.36
Interest = 24,800 − 10,800 = ₹14,000. Time = 14,000 × 100/(10,800 × 15.5) = 14,000/1,674 ≈ 8.36 years.

Q.18 · Answer (B) 33 : 31
For a 20% gain the mixture must cost 75/1.2 = ₹62.50 per kg. By alligation, ₹78 sugar : ₹46 sugar = (62.5 − 46) : (78 − 62.5) = 16.5 : 15.5 = 33 : 31.

Q.23 · Answer (B) 1050 m
The policeman gains 2 km/h, so he closes the 100 m (0.1 km) in 0.05 hours. In that time the thief runs 21 × 0.05 = 1.05 km = 1050 m.

Q.24 · Answer (C) ₹24,000
2A = 4B = 6C gives A : B : C = 1/2 : 1/4 : 1/6 = 6 : 3 : 2. A's share = 6/11 × ₹44,000 = ₹24,000.

English Comprehension

Q.1 · Answer (C) come up with
To 'come up with' a solution is to think of or produce one.

Q.2 · Answer (D) A rainy day
Money saved 'for a rainy day' is kept for a time when it may unexpectedly be needed.

Q.3 · Answer (D) called around but he wasn’t able to
To 'call around' is to telephone several people or places, as when searching for a part; to 'call back' is only to return a call.

Q.4 · Answer (D) Congested
'Barren' means bare or empty; its opposite is 'congested', crowded and full.

Q.5 · Answer (D) Generosity
'Avariciousness' is greed; its opposite is 'generosity'.

Q.6 · Answer (B) Measerement
The correct spelling is 'measurement'.

Q.7 · Answer (A) pride of lions
Related females, cubs and a few adult males make up a 'pride of lions'; cattle have calves, not cubs.

Q.8 · Answer (B) Bohemian
A 'maverick' is an independent-minded, unconventional person, as is a 'bohemian'. The other three are opposites.

Q.9 · Answer (B) move slowly
A comparison with 'than' needs the comparative: 'move more slowly than other domestic animals'.

Q.10 · Answer (A) Atheist
An atheist is a person who does not believe in God.

Q.11 · Answer (D) hole
A 'cavity' is a hollow, a 'hole'.

Q.12 · Answer (C) clash
A 'conflict' of egos is a 'clash' of egos.

Q.13 · Answer (C) QSPR
'I have (Q) not seen (S) her face (P) for a year (R).'

Q.14 · Answer (D) Speak truthfully
To 'call a spade a spade' is to speak plainly and truthfully.

Q.15 · Answer (C) Neccessary
The correct spelling is 'necessary'.

Q.16 · Answer (C) else but your pride which makes
The idiom is 'nothing else but', and 'pride' takes the singular 'makes': 'It is nothing else but your pride which makes you say such a thing.'

Q.17 · Answer (B) BCAD
The taxi stops (B), Mr Satterthwaite runs up the stairs (C), finds the door ajar (A) and pushes it open (D): BCAD.

Q.18 · Answer (A) A best-selling novel has been written by him.
The present perfect 'has written' becomes 'has been written' in the passive.

Q.19 · Answer (A) An inspiring speech was written by him for the event.
The simple past 'wrote' becomes 'was written' in the passive.

Q.20 · Answer (B) very huge
'Colossal' means extremely large.

Q.21 · Answer (A) indifferent
He was failing most of his classes: an 'indifferent' student, though gifted in math.

Q.22 · Answer (B) suggested
Someone 'suggested' that he apply to MIT; he had not even heard of the school.

Q.23 · Answer (A) compare
The acid-tongued professor was known to 'compare' his audience to a herd of cows.

Q.24 · Answer (B) introversion
Part of his isolation came from his own 'introversion'; the rest, the passage says, was cultural.

Q.25 · Answer (B) exclude
No one had tried to 'exclude' him, yet he felt alone.

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